CSM V3 — ASSEMBLY PROCEDURE

Wood Base V1.1 Electrical Mounting • Rev V1.0 • ~4-6 h • Page 1: Drilling + Mechanical

BUILDER: ______________________ **DATE:** ______________________

PHASE A — PREPARATION (30 min)

- [ ] **1.** Print drilling_guide.pdf at 1:1 scale. Verify with ruler that 50 mm marks measure 50 mm.
- [ ] **2.** Gather all hardware per hardware_order_sheet.pdf checklist.
- [ ] **3.** Stage tools: drill, drill bits (Ø2 / Ø3.2 / Ø3.5 / Ø4.5 / Ø5.5 / Ø100 saw), screwdrivers, wrenches, calipers.
- [ ] **4.** Wood-base orientation: long edge = X (500 mm), short edge = Y (400 mm). Mark "FRONT" arrow on -Y edge.
- [ ] **5.** Mark center of wood base (datum 0, 0). Draw faint +X and +Y axis lines with pencil.
- [ ] **6.** Tape the drilling_guide printout to the wood with axis lines aligned to the printed datum.

PHASE B — DRILLING (1 h)

- [ ] **1.** Center-punch ALL 21 hole positions through the paper template into the wood.
- [ ] **2.** Remove paper template. Pilot-drill every position with Ø 2.0 mm bit.
- [ ] **3.** Drill 10 × Ø 3.2 mm holes (Mega ×4, Buck#1 ×2, Buck#2 ×2, Terminal ×2).
- [ ] **4.** Drill 2 × Ø 3.5 mm holes (TB6600 flange).
- [ ] **5.** Drill 4 × Ø 4.5 mm holes (PSU chassis).
- [ ] **6.** Drill 4 × Ø 5.5 mm holes (NEMA 23 motor flange).
- [ ] **7.** Cut Ø 100 mm take-down opening with hole saw or jigsaw, centered on datum.
- [ ] **8.** (Optional) Counterbore PSU + Motor holes Ø 8 × 2 mm from UNDERSIDE to recess nuts.
- [ ] **9.** Vacuum dust. Verify hole positions with calipers — tolerance ±1 mm.

PHASE C — MECHANICAL MOUNTING (1.5 h)

- [ ] **1.** Dry-fit PSU at (-195, 0). Verify all 4 chassis screws align with Ø 4.5 holes.
- [ ] **2.** Bolt PSU: M4 × 30 + washer top, washer + nut bottom. Tighten cross-pattern.
- [ ] **3.** Set NEMA 23 + HG5 motor at (+85, -47), shaft pointing UP. Verify gearbox orientation.
- [ ] **4.** Bolt motor: M5 × 35 + washer + lock washer + nut. Apply Loctite 243 on threads.
- [ ] **5.** Mount TB6600 at (+200, -47). Bolt: M3 × 25 + washer + nut. Terminals face +X (edge).
- [ ] **6.** Install M3 × 10 mm nylon standoffs at the 4 Mega positions (+200, +75 area).
- [ ] **7.** Mount Arduino Mega on standoffs. Bolt: M3 × 30 + washer + nut. USB port faces +X.
- [ ] **8.** Mount LM2596 Buck #1 at (-50, +160) using M3 × 22 + 5 mm standoff (or 3M VHB).
- [ ] **9.** Mount LM2596 Buck #2 at (+50, +160) same method as Buck #1.
- [ ] **10.** Mount 24V terminal block at (0, +180). Use M3 × 22 OR DIN-rail clip.
- [ ] **11.** Build operator panel per operator_panel.pdf. L-bracket to FRONT (-Y) edge of wood base.

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PHASE D — WIRING (2-3 h, see CSM_V3_WIRING_V1_0.md)

- [ ] **1.** Wire mains FIRST with PSU DISCONNECTED from wall. 16 AWG brown/blue/g-y.
- [ ] **2.** Order: wall → E-stop → 2 A T-fuse → PSU L/N/PE terminals.
- [ ] **3.** Wire 24 V DC bus: PSU +V/-V → 24V terminal block. 14 AWG with ferrules.
- [ ] **4.** Branch 24 V to TB6600 power, Buck #1 IN, Buck #2 IN.
- [ ] **5.** BEFORE downstream loads: power bucks alone, adjust each to 5.10 V no-load.
- [ ] **6.** Wire 5 V #1 → Mega VIN + GND + 6× servo headers. Single-point GND.
- [ ] **7.** Wire 5 V #2 → Pi GPIO 5 V + GND (or USB-C wall wart for bench).
- [ ] **8.** Wire Mega → TB6600 step/dir/en (D5/D6/D7 with 220 Ω each).
- [ ] **9.** Wire NEMA 23 4-wire to TB6600 A+/A-/B+/B- (check motor color code).
- [ ] **10.** Wire E-stop second NC contact → Mega D2 with 10 kΩ pull-up.
- [ ] **11.** Cable-tie all wiring per ELECTRICAL_LAYOUT §2.6 (mains FRONT, 24V BACK).
- [ ] **12.** Label every wire bundle with tape: AC / 24V / 5V / SIG.

⚠ DO NOT CONNECT MAINS UNTIL BENCH-TEST PHASE 1 PASSES ⚠

PHASE E — VERIFY BEFORE FIRST POWER (15 min)

- [ ] **1.** Re-check every bolt is tight; no loose washers visible.
- [ ] **2.** Confirm all 4 motor bolts have lock washers + Loctite 243.
- [ ] **3.** Multimeter continuity: PE wire wall plug → PSU FG. Must be < 1 Ω.
- [ ] **4.** Multimeter resistance: PSU +V to -V terminals UNPOWERED. Should be high (>1 kΩ).
- [ ] **5.** No bare conductor exposed at any mains terminal. Cap unused terminals.
- [ ] **6.** Verify wire colors: brown=L, blue=N, green/yellow=PE, red=+, black=GND.
- [ ] **7.** PSU voltage selector SWITCH set to match your mains (115 OR 230 V).
- [ ] **8.** Photograph the finished assembly (top + 4 sides) for the record.
- [ ] **9.** Proceed to BENCH-TEST CHECKLIST (bench_test_checklist.pdf).

NOTES / DEVIATIONS: